Sales Forecasting & Demand Intelligence

Prepared for the Head of Supply Chain and the CFO | July 2026

Executive Summary

Looking at the last four years of sales, the business has shown steady growth, although demand changes throughout the year. Sales regularly increase during the holiday season and again in March, while a few weeks stand out with unusually high or low performance. Based on these patterns, a forecasting and monitoring system was developed to estimate sales for the next three months, identify unusual sales activity, and classify products into demand groups. Overall, the business is expected to continue growing. Technology currently contributes the highest revenue, although its growth may slow, while a relatively small group of high-performing products should receive priority in inventory planning.

Key Findings

- **Technology leads revenue** with $827,456 in total sales, ahead of Furniture ($728,659) and Office Supplies ($705,422).
- **The East region grows the most reliably** — its year-over-year growth barely wavers, making it the safest region to plan around, while the South region swings the most.
- **Shipping is consistent** across all regions, averaging about 4 days from order to shipment, so logistics is not currently a source of regional risk.
- **Sales are seasonal and repeat every year**: a dip in February, a spike in March, and the strongest period from September through December, peaking around Black Friday and the holidays.
- **Three forecasting methods were tested**; the table below shows how they compared on the most recent 12 months of held-out data.

Forecasting Method	Typical Error	Verdict	
Prophet (selected)	15.7%	Most accurate — used for the forecast above	
XGBoost	25.2%	Less reliable with only 4 years of history	
SARIMA	27.8%	Struggled with volatile months	

3-Month Sales Forecast (2019 Outlook)

Using our best-performing model together with the historical seasonal pattern and current category growth trends, we project total company sales for the next three months below. The “low” and “high” columns reflect the model's typical margin of error (about ±16%), so treat the expected figure as the most likely outcome and the range as reasonable bounds for planning.

Month	Expected Sales	Low Estimate	High Estimate
January 2019	$44,500	$37,500	$51,500
February 2019	$20,400	$17,200	$23,600
March 2019	$60,300	$50,800	$69,800
Q1 2019 Total	$125,200	$105,500	$145,000

In plain terms: expect a normal, modest January, a noticeably quieter February (this is a recurring pattern, not a warning sign), and a strong rebound in March that has occurred in every year of available data.

Top 3 Anomalies Detected

An automated scan of weekly sales flagged 11 unusual weeks over four years. The three recurring patterns below account for the large majority of them.

Period	What Happened	Likely Cause
Late Nov – Dec	Sharp, repeated spikes in weekly sales, the largest of the year	Black Friday / Cyber Monday promotions and year-end holiday buying
March	A recurring, unusually large spike appearing in all four years	End-of-quarter corporate purchasing and spring promotions
Early January	Sales drop well below the normal range	Post-holiday slowdown as customer spending resets

Product Demand Segmentation & Stocking Strategy

We grouped all 17 product sub-categories into four segments based on how much they sell, how fast they're growing, and how volatile (unpredictable) their sales are. Each segment calls for a different stocking approach.

Segment	Example Products	Behavior	Stocking Strategy
Core Growth (highest revenue)	Binders, Chairs, Phones, Storage, Tables	Highest total sales; strong 22% growth	Automated reorder with expanded buffers
Rising Stars	Accessories, Appliances, Art, Labels, Paper	Smaller today, but fastest-growing (43%)	Increase buffers; protect against stockouts
Niche / High-Value	Copiers, Machines	Fewer, pricier orders; most unpredictable	Just-in-time ordering; avoid overstocking
Declining	Bookcases, Envelopes, Fasteners, Furnishings, Supplies	Lowest sales; slightly shrinking	Markdown, clear out, reduce reorders

Business Recommendations

- **1. Build inventory buffers 25–30% ahead of Black Friday/holiday season and again ahead of March.** These two windows account for the largest and most consistent sales spikes in four straight years of data; running lean during these weeks risks lost sales at the highest-demand moments.
- **2. Apply the segmented stocking strategy above, starting with the four highest-impact sub-categories.** Binders, Chairs, Phones, Storage, and Tables together generate the most revenue and are still growing 22% a year — they deserve automated reordering and larger safety buffers. Bookcases, Envelopes, Fasteners, Furnishings, and Supplies are flat to declining and are tying up capital that could be redeployed; begin phasing them down with markdowns.
- **3. Prioritize the East region for growth investment and treat the Technology category's 2019 outlook with caution.** East shows the steadiest year-over-year growth of any region, while Technology — today's top revenue category — is projected to decline roughly 8% next year even as Furniture (+9%) and Office Supplies (+8%) grow; a proactive review of Technology pricing or promotions is warranted before the dip materializes.

Risk & Limitation

This system is built on four years of historical data and, like any forecast, it describes patterns — it does not know about future events such as new competitors, supply disruptions, or promotions that haven't happened yet. Its typical margin of error is about 16%, and two independent methods used to flag “unusual” sales weeks agreed on which weeks were unusual less often than expected, meaning some judgment should still be applied before treating any single alert as confirmed. We recommend treating this system as a decision-support tool that narrows down where to focus attention, not as a fully automated replacement for experienced planning judgment.